Digital
Transformation
Research Programme

Maritime Autonomous Surface Ships (MASS)

Creating a framework for efficiency, safety and compliance



MOL
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1. Introduction

The maritime industry, in line with many other industrial and transport domains, is undergoing a significant and unparalleled technological change driven by increased automation and digitalisation. One of the most direct examples of this trend is the ongoing development of Maritime Autonomous Surface Ships (MASS) – a paradigm shift with the potential to significantly impact many of the critical issues faced by the sector today.

The technologies involved offer opportunities to generate new levels of operational efficiency, delivering significant sustainability and fuel consumption gains. In addition, safety and reliability may be significantly improved by reducing the high percentage of maritime accidents caused by human error.

MASS can also assist the industry in meeting the growing international challenge of crew shortages and the management of seafarer welfare and workload, reconfiguring vessel operational structures through the integration of autonomous technologies to better support those onboard in performing their duties.

At its core, the MASS concept offers the capability to operate vessels partially or fully autonomously, reducing the burden and mitigating safety risks for the humans involved. Analysis of investigated navigation accidents has shown that approximately 78% of cases are linked to "human action"[1], an issue that will be directly impacted by improvements in automation – though it is important to guard against the introduction of new forms of error in both the design and assessment of MASS while attempting to minimise our human faults.

Fundamentally, commercial applications for MASS can be broken down into three categories – Remote Control, Bridge Navigation Assist and Full Autonomy. These different capabilities utilise various combinations of hardware and software in the form of sensors (primarily radar, cameras and lidar), artificial intelligence, data processing and collision avoidance algorithms, as well as dynamic situational awareness and route planning applications. Depending on the autonomy level in question, the systems will be designed to recommend or deliver optimised safe transit routes in real-time.

Bridge Navigation Assist systems are used to augment traditional navigation processes and provide support to crews, delivering heightened situational awareness and a wider range of data to improve decision making. This enables personnel to maintain overwatch while concurrently undertaking wider bridge duties, reducing the overall workload and improving both vessel safety and crew welfare. At the highest degrees of autonomy, the crew can be freed to either undertake other tasks or potentially be completely removed from performing the task role. Machine learning, computer vision, and control systems can be combined so that the vessels themselves have the capability to execute operational tasks and decisions autonomously.

Whichever model is employed, the human will remain central to how operations are carried out – both ethically and technically. To cater for these scenarios, appropriate communication technologies and protocols are needed, with suitable bandwidth communication (i.e. hardware and connectivity) to support the functionality required. The development of Remote Operation Centres (ROCs) and the changing human roles and responsibilities that these facilities will create will open up a new phase for the maritime industry, leading to changes that must be fully considered as innovation progresses.

Bridge Navigation Assist and Full Autonomy are, in effect, a collection of technologies. When combined with the human, they become 'capabilities'. It is this holistic model that offers the opportunity to radically change and improve operations in the maritime domain.

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IMO's four degrees of autonomous operation

Maritime Autonomous Surface Ships (MASS) can be designed to operate at varying degrees of independence from human input or intervention. The IMO (International Maritime Organisation) categorises four degrees of maritime autonomy:

- **Degree one – Ship with automated processes and decision support:** Seafarers are on board to operate and control shipboard systems and functions. Some operations may be automated and at times be unsupervised but with seafarers on board ready to take control.
- **Degree two – Remotely controlled ship with seafarers on board:** The ship is controlled and operated from another location. Seafarers are available on board to take control and to operate the shipboard systems and functions.
- **Degree three – Remotely controlled ship without seafarers on board:** The ship is controlled and operated from another location. There are no seafarers on board.
- **Degree four – Fully autonomous ship:** The operating system of the ship is able to make decisions and determine actions by itself.

1.1. The compliance imperative

Regulatory compliance is one of the central pillars of creating trust and confidence in MASS across the full range of stakeholders. It is not only a legal necessity – trust is developed when stakeholders can see that appropriate risk management and mitigation measures have been applied in a structured way. Furthermore, within the globally interconnected marine environment of both shipping and port operations, interoperability with the existing maritime infrastructure based on compliance with regulatory frameworks is a key facet of safe and efficient international trade.

Across all autonomy levels, it is critical that MASS demonstrate that they adhere to international regulations and safety standards, such as the upcoming International Maritime Organisation (IMO) MASS Code, the International Regulations for Preventing Collisions at Sea 1972 (COLREGs), the International Convention on Standards of Training, Certification and Watchkeeping for Seafarers (STCW) and the International Convention for the Safety of Life at Sea (SOLAS).

Beyond safety, these tools help to define legal responsibilities and liabilities in the event of accidents or disputes, providing clarity that will be essential for insurance purposes and broader accountability. Each is examined in more detail later in this paper, in the context of their impact on MASS operations.

The human-centric nature of these documents remains a critical challenge to the commercial adoption of MASS and its safety, economic and environmental benefits. Developments in autonomy are forcing nations and various industrial sectors to adapt their implementation of different regulations, and maritime would similarly benefit from having agile, robust and flexible frameworks in place to address the evolving nature of the sector. Collaboration between governments, industry stakeholders, and regulatory bodies will be key in addressing these challenges and providing a streamlined, evidence-based pathway.

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1.2. Challenges faced in MASS introduction

The purpose of this paper is to address the challenge of ensuring the safe operation of MASS, which will require the industry to comply with regulations while also developing maritime autonomous hardware and software technology systems capable of properly executing operational procedures.

The paper intends to raise awareness of the steps required to adapt the COLREGs, STCW and SOLAS regimes to the novel requirements of autonomous systems in maritime and highlight the need for multi-stakeholder collaboration to accelerate this process. This will require Governments, as funding partners, to work with Regulators and their Recognised Organisations, alongside Industry actors including Ship Operators, Ports, and other stakeholders, to meet this challenge.

A collaborative effort will not only help to accelerate the adaptation of regulatory regimes but will also assist in ensuring that operational safety standards align with the emergent capabilities of autonomous technologies. Such activities will in turn facilitate commercial adoption and distribution of the benefits of MASS to all stakeholders involved.

In order to create the awareness and agreement necessary to deliver the level of collaboration called upon in this paper, the following sections aim to define the critical importance of incorporating applied human intelligence into the fabric of MASS. The human should always remain at the centre of autonomy.

Before technological advancement can proceed, there are essential key steps that must be taken to ensure that MASS development follows a holistic and human-centric approach, to foster safer, more reliable, more intuitive and socially acceptable systems. The operational technologies required for MASS must be proven to operate with a high level of safety as we continue the development of competency in all stakeholder roles so that the human can monitor or intervene in autonomous operations in the event of failure.



By fostering collaboration, we can merge the learnings from ‘applied human intelligence’ (i.e. the competency of human crews) with ‘artificial intelligence-driven software’ that can apply existing knowledge and insights in new and better ways.

Despite the remarkable strides made in AI, the complexities of human cognition and decision-making are yet to be fully replicated by the technology. This apparent gap stems from various factors, reflecting the complexity of translating human intuition into algorithmic frameworks. Properly incorporating the complexity of human behaviour into autonomous models poses a range of challenges to the widespread adoption of MASS – these issues will be explored below.

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2. Technological evolution

Transitioning from human-operated systems to augmented or autonomous operations will require a series of critical areas to be addressed, in terms of regulatory frameworks, liability, and legal responsibility.

Human operators bring inherent value to operations, including perception, interpretation, and the ability to adapt to dynamic environments. However, their limitations, such as attention span, error and fatigue, must be acknowledged to identify areas where automation can improve performance and safety.

Understanding human factors like cognitive load and proneness to error is crucial in ensuring that autonomy enhances rather than diminishes safety. This will involve analysis in areas including decision-making processes, human limitations, interpersonal dynamics, communication protocols, crew interactions, training, and the division of responsibility among human operators. Ethical and legal considerations, such as safety, accountability, and job displacement, must also be addressed to mitigate associated risks.



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These considerations take on added importance in industries facing personnel shortages, like the maritime sector. Cultivating trust in autonomy requires transparent communication, rigorous testing, and a proven track record of reliability and safety.

Ultimately, ensuring safe optimisation of maritime operations requires a balanced understanding of human factors, training, and risk management. Adopting a systems engineering methodology – a transdisciplinary and integrative approach applying systems principles and concepts[2] – will be crucial in deconstructing and reallocating tasks effectively, prioritising safety and human well-being.

Advancements in engineering, data science, computing technology and model-based tools have created a platform for the development of autonomous systems in the maritime industry. These AI-enabled systems can observe, analyse, make decisions and take action independently of the engineers and crew on board vessels. Commercialisation[3] of autonomous systems is contingent on stakeholders such as developers, vessel owners, operators, regulators and insurers being confident in the overall proficiency and performance of the technology, recognising its potential benefit as a tool to support human needs rather than replace the human entirely.

Ships operate under dynamic internal conditions and within a constantly changing external environment that presents multiple hazards impacting safety. For MASS to be effective it must combine the 'applied human intelligence' of the crew with AI software to achieve operational safety and performance that cannot be attained by humans or AI technology independently. Substantial evidence of the system producing correct, timely and dependable analysis, delivering decision-making and operational actions in line with the design intent, including resiliency when dealing with adverse situations (i.e. edge cases), will be needed to demonstrate its worth.

Researchers will require the expertise of crews with intimate knowledge of real-world maritime and bridge environments, as well as personal insights into operator behaviours and needs, to recognise the nuanced aspects of human interaction with vessels, infrastructure and other stakeholders. This will be necessary in identifying the 'pain points', inefficiencies, and opportunities for improvement experienced during actual ship operations. Understanding these issues will enable the design of MASS that not only meet functional requirements but also enhance usability, safety, and overall operability.

Combining this process with a maritime-centric Operational Design Domain (ODD) will help to determine the conditions under which a MASS can safely operate, specifying its environmental, geographical, and operational constraints. Defining these parameters within an ODD framework will help to ensure that MASS are only deployed in situations where their performance can be reliably assured to enhance overall safety and reliability.

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3. Adapting regulatory frameworks

The advent of autonomous vehicles promises transformative changes across a variety of domains; however, this technological revolution has not happened without its share of difficulties, with the full potential of these systems intimately tied to robust regulatory frameworks that ensure and enable their safe integration. While this paper focuses on MASS and the maritime sector, it is essential to understand the broader context of autonomous vehicle regulations across all domains and contextualise the challenges inherent in their implementation.

Autonomous vehicle regulation is a complex and evolving landscape, characterised by a patchwork of laws and standards that vary significantly across different jurisdictions. For land-based Connected Autonomous Vehicles (CAV), regulatory frameworks have been changing rapidly but face significant hurdles due to the fact that development has focused on technology demonstration rather than the assurance of the autonomous system itself. Different regions adopt varying approaches, ranging from stringent regulations to more permissive environments, creating a fragmented landscape.

Challenges arise in areas such as liability, data privacy, and infrastructure compatibility. One key element is the development of standards for vehicle-to-everything (V2X) communication protocols, essential for CAV coordination and safety. Additionally, the creation of proper definitions of liability and insurance frameworks that will apply in the event of accidents involving CAVs remains contentious, while ensuring data security and privacy protection in situations where data sharing is necessary for CAV functionality poses another hurdle.

International harmonisation efforts face obstacles due to divergent national priorities and regulatory philosophies, which means that achieving consensus on global standards remains a formidable challenge. Fortunately, the structure of the maritime sector, with IMO as the global regulatory authority, obviates several of these structural issues.

Moving from land to the skies, the aviation industry has similarly been exploring the potential of autonomous aerial vehicles (AAVs) for applications like delivery and urban air mobility. However, navigating the regulatory airspace has proven complex due to safety concerns over the integration of AAVs with traditional manned aircraft. Regulatory bodies must develop standards for airworthiness, airspace management, and pilot certification for autonomous systems, presenting significant challenges in ensuring the safety and reliability of these vehicles. While the International Civil Aviation Organisation (ICAO) has issued standards and recommended practices for the certification and operation of autonomous aircraft, the level of implementation of these standards varies among member states, leading to discrepancies in regulatory frameworks.

The maritime sector's regulatory and third-party assurance structure is unique. As mentioned above, the IMO plays a crucial role in developing overarching goal-based frameworks that provide a foundation for national regulatory bodies to adapt and implement regulations tailored to their specific maritime industries and legal frameworks.

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Japan, the UK and Singapore each have distinct approaches to regulating maritime autonomy. Japan has been proactive in fostering innovation and collaboration between industry stakeholders and government agencies, establishing dedicated research centres and initiatives to promote autonomy, particularly the Design of the Future of Full Autonomous Ships (DFFAS) programme, which has just entered its second phase. South Korea too will enter their second phase, with the goal of the Korea Autonomous Surface Ship (KASS) project aiming to successfully develop core technology and inform the wider needs for MASS.

The UK has adopted a flexible and adaptive risk-based assessment approach, promoting innovation and close collaboration with industry stakeholders. It has focused on creating a favourable regulatory environment through initiatives. The Maritime and Coastguard Agency (MCA) has also drafted guidance of its own, under MGN 664 and WBC3.

Singapore, as a major maritime hub, has prioritised the development of autonomous shipping within the Maritime and Port Authority of Singapore's (MPA) 'Smart Port Challenge', aiming to leverage advanced technologies for enhanced efficiency while upholding safety standards. Overall, while these countries share common goals of promoting maritime autonomy, their regulatory approaches reflect their unique priorities, capabilities, and challenges. To succeed they will all need to embrace cooperation and align with global standards, which will remain essential to ensure the effective integration and interoperability of MASS operating across territorial waters.

It is clear that the rapid pace of technological innovation has outpaced the respective regulatory processes across all domains, leading to regulatory lag. The inherent complexity of these new technologies makes it difficult for regulators to establish clear guidelines for safety and performance standards, but new frameworks should make sure to prioritise flexibility and adaptability to accommodate the evolving nature of autonomous systems.

3.1. Legal and ethical questions in MASS development

The legal landscape surrounding MASS is a complex, rapidly evolving web of international maritime law, domestic regulations, and emerging frameworks. The intersection of traditional maritime law with emerging technological advancements raises new types of legal concerns to which the sector is unaccustomed. Traditional maritime legal frameworks typically rely on 'case law' and precedents that have already been set – in a novel and rapidly changing technological landscape these frameworks will need to be adapted to adequately address various aspects of MASS development, including safety, liability, environmental protection, and navigational standards.

At the international level, the United Nations Convention on the Law of the Sea (UNCLOS) establishes the rights and responsibilities of nations. Predating MASS development, this convention poses a challenge as to its applicability in an autonomous context – if unchanged, the legal responsibilities of different parties under UNCLOS following an incident involving MASS would be unclear.

The complexities of flag/port/coastal state relationships, particularly regarding the right to freedom of navigation and the rights of coastal states to approve passage, need to be clarified.

The legal framework for MASS is primarily governed by the International Maritime Organisation (IMO), the specialised agency of the United Nations responsible for regulating shipping.

The IMO's Convention on the International Regulations for Preventing Collisions at Sea (COLREGs) establishes rules to prevent collisions between vessels, currently assumed to be manned vessels. Ratification of a non-mandatory MASS Code is being sought within the IMO Maritime Safety Committee (MSC) 109, sitting in December 2024 (described below).

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International Maritime Organisation (IMO) MASS Code

In April 2022, IMO MSC 105 defined a roadmap to develop a goal-based MASS Code with a view to future adoption. This is scheduled to be released as a non-mandatory Code on 1st January 2025 (for immediate use). Later on in May 2024, MSC 108 has concluded that the Code's formal adoption for mandatory implementation would be expected not before 1st January 2032.

The Code is goal-based and will only contain high-level goals and functional objectives, to remain independent of any technology bias. However, without a definition of the technology and operating models to be deployed, and with no clear description of how these should be assured and verified, it will instead be left to Industry, Member States and Class Societies to create methods to determine suitability. The Code will only provide a framework against which a vessel using autonomous technologies can be assessed and certified for commercial operations within IMO's structure – it will not provide the means for how this can be done.

The process of agreeing and drafting the MASS Code has been a challenging one. The complexity of crafting a universally acceptable regulation navigating diverse interests and perspectives within the community is exemplified by its goal-based nature and lack of defined methods for assurance and verification. The issues were multifaceted, particularly in achieving consensus among IMO member states with varying interests and priorities. Whilst constraining content, it is apparent that attempts to reconcile these diverse perspectives and interests among member states will need to continue until at least the planned adoption period.

It is to be further noted that to date none of the MSC Sub-Committees, including HTW, (Human Element, Training and Watchkeeping), has begun any work relating to outputs associated with the MASS Code, for example developing a competence standard for Remote Operators. This is primarily related to the fact that IMO is waiting for work on the MASS Code to be completed so that work packages can then be formed for the Sub-Committees based on the 'demand signals' created by the Code.

National legislation plays a significant role in regulating MASS operations within their territorial waters and exclusive economic zones. These frameworks typically cover vessel certification, operational requirements, and liability regimes. However, there is a lack of uniformity among national regulations, potentially leading to legal ambiguities and conflicts, particularly in cases involving operation across national boundaries.

Liability remains a complex issue, particularly when it comes to accidents or incidents involving MASS. Determining responsibility and liability may require input from various stakeholders, including vessel operators, manufacturers, software developers, and regulatory authorities. Traditional concepts of maritime law, such as the doctrine of 'personal fault' or 'privity' (an interest in a transaction, contract, or legal action) may need re-evaluation to ensure their applicability in the context of autonomous technology.

In the UK, 'The Future of Transport Regulatory Review' included a focus on Maritime Autonomy and Remote Operations, seeking to overhaul transport regulation to accommodate the evolving landscape. Conducted by the Department for Transport (DfT) between September 28 and November 22, 2021, it aimed to gather insights to support the development of a robust regulatory framework for autonomous and remotely operated ships through primary legislation.

The report's findings indicated broad support for defining MASS and Remote Operations, modifying the Master's responsibilities, and regulating Remote Operation Centres (ROCs). Responses were more varied when evaluating whether current powers granted to harbours and ports would be sufficient in the MASS era, and whether additional legislation on marine security and liability was needed. Challenges in relation to insurance and compliance with international standards were also highlighted in the report.

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Moving forward, the UK government plans to undertake comprehensive regulatory updates to support MASS under primary legislation, however this is dependent on parliamentary time being available, and continued stakeholder engagement for secondary legislation development.

Ethics in MASS is another topic for debate, as stakeholders seek to determine how best to ensure the safety of crew, passengers, and other vessels, while also prioritising environmental protection and maintaining compliance with international maritime laws and regulations. Vessels' algorithms must be programmed to navigate with collision avoidance parameters that minimise risks to human life and respect established maritime conventions, norms and procedures. Moreover, mechanisms for transparency and accountability must be built into the design and operation of MASS to enable meaningful human oversight and intervention when necessary.

Additionally, potential job displacement and clarity as to how the expansion of autonomous systems may reposition maritime roles into ROCs rather than on board require ethical consideration. By incorporating these concerns from the outset at the design stage, potential risks can be mitigated and a foundation laid to build stakeholder and public trust in autonomous maritime technologies. International cooperation will be crucial in that regard.

In summary, regulations have struggled to keep pace with current technological developments. The legal landscape for MASS is multifaceted, encompassing international treaties, national legislation, safety standards, liability frameworks and environmental regulations. As these systems continue to advance towards commercialisation, domestic legislation will need to be tailored to handle the unique challenges involved, with stakeholders collaborating on the development of comprehensive and harmonised legal frameworks that support the safe, efficient, and sustainable integration of MASS into the global maritime industry.

Perhaps one good example to learn from is the recent development of global AI regulatory management, with the introduction of the EU AI act. This legal framework will unify how AI is regulated across the single market of the 27 EU Member States, carrying significant international implications as it covers all AI systems affecting people in the EU regardless of where systems are developed or deployed.

The compliance requirements are largely determined by the level of risk the usage of an AI system poses to people's safety, security, or fundamental rights, applying a tiered compliance framework with obligations across the entire AI value chain. Most requirements fall upon the developers and deployers of AI systems classified as 'high-risk', and on general-purpose AI systems (including foundational models and generative AI systems) posing 'systemic risks.'

AI applied in the transportation industry – including maritime – is classified as 'high-risk' due to the safety-critical nature of operations. Pre-market 'conformity assessments' (verification and validation) will be required for all high-risk AI systems to demonstrate their compliance with harmonised standards (under development) before being placed on the market. Noncompliance could leave the offending company at risk of fines of up to €15 million or 3% of total worldwide annual revenue (whichever is higher).[4]

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4. Regulatory review – COLREGs, STCW and SOLAS

Robust engineering requirements form the cornerstone of any successful project, serving as the 'blueprint' for its entire lifecycle. They provide a pathway for clarity, consistency and alignment on direction, reducing rework and mitigating risks. The ability to track, document and link requirements throughout the project is also vital, to enable stakeholders to understand the rationale behind design decisions, facilitate change management, and help ensure compliance with regulations and standards.

The interrelationship between requirements and architecture is important for successful system development. Requirements serve as the foundation, whereas the architecture the project is built on allows components to interact to meet those requirements – translating the requirements into functional designs, where each requirement aligns with a function within the architecture, ensuring traceability and supporting validation. This can be further improved through an iterative process of refinement so that the architecture evolves responsively, integrating feedback and optimising its alignment with the requirements.

The challenge of applying this approach in the development of maritime autonomy is linked to the maturity of currently identified requirements, which need to be further refined. The human-centric nature of COLREGs, STCW and to some extent SOLAS currently act as inhibitors in this regard.

4.1. COLREGs

The International Regulations for Preventing Collisions at Sea (COLREGs) serve as the cornerstone for safe navigation. The modern convention was drafted with an understandably human-centric focus by the IMO in 1972 and provides a framework governing the conduct of vessels to prevent collisions and ensure the safety of navigation in all waters (up to the first bridge in navigable waterways). Incorporating rules for vessel lights, sound signals, right of way, and other essential parameters, COLREGs are a vital tool in reducing the risk of accidents for mariners worldwide, promoting clear communication and harmonised practices. In reality however, application of the regulations requires specific issues to be interpreted on a case-by-case basis to determine how multiple rules are concurrently considered, prioritised and/or applied.

Despite the precision of the regulations, the unpredictable nature of maritime environments demands human interpretation, which will vary in accordance with several factors, such as training and experience. Vessels encounter diverse situations, from adverse weather conditions to unexpected encounters with other ships or hazards, and the human-centric approach to interpreting COLREGs requires mariners to navigate the fine line between adhering strictly to the rules and exercising prudent judgment in real-time dynamic scenarios. It is these judgments that truly safeguard lives and allow for safe navigation at sea.

It is beyond the scope of this paper to define all issues that could arise with the language and terminology found in COLREGs in the context of MASS operation – that requires separate detailed examination for each specific level of autonomy and each specific COLREGs rule. However, three specific and significant rules are examined below.

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Adapting COLREGs for autonomous operations – Responsibility, Look-out and Good Seamanship

COLREGs Rule 2 (Responsibility) raises a number of issues in relation to MASS operation. It has two sections, with Section (a) stating *"Nothing in these Rules shall exonerate any vessel, or the owner, master or crew thereof, from the consequences of any neglect to comply with these Rules or of the neglect of any precautions which may be required by the ordinary practice of seamen, or by the special circumstances of the case."*[5] The language regarding 'good seamanship' and 'prudent judgment' leaves room for interpretation, as they are not specifically defined.

Section (b) states that *"In construing and complying with these Rules due regard shall be had to all dangers of navigation and collision and to any special circumstances, including the limitations of the vessels involved, which may make a departure from these Rules necessary to avoid immediate danger."*

Both sections of Rule 2 create a duty of care based on the knowledge, skills and experience of the seafarer. Ships need to take the best action to avoid collisions even if they find themselves in special situations that cannot be covered by basic navigation alone. Consequently, many aspects of ship operation are left to the judgment of the operator based on practices that have evolved over many years. There is no written description of how to apply these learned judgment skills as these are developed through experience, training and competence – the pillars of a human-centric approach.

It is imperative that, when utilised, autonomous functions can safely respond to these kids of circumstances despite this ambiguity. MASS will require self-identification capabilities to be able to define whether the vessel is operating within or outside of its Operational Design Domain (ODD) – the operating environment for which the system is designed – as well as the capacity to notify personnel such as Officers of the Watch (OOW) accordingly.

Determining the "Master" or responsible party becomes a more complicated process, as traditional maritime roles may not directly translate to autonomous operations and some roles may transfer to an ROC. Ensuring effective communication between MASS and traditional manned vessels to make sure the principles of safe navigation and collision avoidance are collectively adhered to will be paramount. Additionally, the decentralised nature of autonomous systems without direct human intervention introduces complexities in assigning responsibility. Determining whether responsibility lies with the ship's owner, manufacturer, or operator, and how liability is allocated in case of collision or non-compliance, becomes a more complicated issue when considering the distributed nature of decision-making systems and the impact of potential malfunctions. Such issues require clarification.

COLREGs Rule 5 states: *"Every vessel shall at all times maintain a proper look-out by sight and hearing … in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision."* This terminology is clearly human-centric but may be interpreted as the situational awareness of whomever controls the ship and their ability to take informed decisions to avoid collisions and other incidents. However, clarity on how to accomplish this role is required. Moreover, the responsibility for maintaining a lookout is less clear when there are no human operators on board to oversee the vessel's navigation.

COLREGs Rule 8 outlines the actions to be taken to avoid collisions, including alterations of course and speed. However, the term "to avoid collision" lacks a precise definition, leaving room for subjective interpretation. While this rule provides general guidance, its application of the term "early and substantial action" also lacks quantifiable parameters and raises questions about how MASS vessels should prioritise collision avoidance manoeuvres and interact with other vessels. Factors such as vessel size, speed, and manoeuvrability can influence what actions constitute adequate collision avoidance, introducing ambiguity into maritime operations.

Consideration should also be given to defining the term 'good seamanship', as well as 'timely', 'too close', 'safe distance', 'vessel is clear', and 'sufficient distance'. These are often referred to in case law and have therefore arguably been assigned meaning, but they have not been codified. While 'good seamanship' has been described as "appropriate practice for the operation of ships", there is no specific definition. These ambiguities could pose challenges during complex and dynamic situations.

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4.2. STCW and the role of the crew

The purpose of the International Convention on Standards of Training, Certification and Watchkeeping for Seafarers (STCW), last updated at MSC102 in 2020, is to ensure that seafarers possess the necessary skills, knowledge and competence to operate vessels and respond effectively to emergencies at sea. The STCW Convention and Code lay down minimum standards for the training and certification of seafarers globally. While the Convention addresses general principles, the detailed requirements are outlined in the Code. However, there are no quantitative standards, and certification of qualifications is left to IMO Member States, who then delegate that responsibility to educational and training institutions.

Crew interaction involves a blend of technical protocols, communication procedures, and interpersonal dynamics that combine to allow safe navigation and operations to be performed. Mariners must adhere to established communication protocols using standardised terminology and procedures, but must also clearly communicate navigational intentions, updates on vessel status, and coordination of manoeuvres via various means, including radio communication, signalling devices, and onboard communication systems.

Each crew member is typically trained to operate specific equipment and systems relevant to their roles on board, such as navigation instruments, engine controls, and communication devices, necessitating teamwork and collaboration during routine tasks, as well as emergency situations. They are trained to work together seamlessly, following predefined roles and responsibilities to execute manoeuvres, conduct safety drills, and respond to emergencies like fires, collisions, or medical incidents. This teamwork relies on trust, clear communication and familiarity with each other's capabilities.

For MASS, we need to define how – and potentially where – these navigational roles will be augmented or transferred, identifying which competencies can be replaced by MASS technologies and which cannot. Furthermore, the combination of the human and these new technologies is likely to change the nature of certain onboard roles and may therefore affect the competencies required of the human as the role evolves.

The STCW also plays a crucial role in determining the appropriate manning levels for modern vessels and ensuring that the crew on board possess the specialised skills required for safe and efficient operations. The need for specialist skills is particularly significant due to the limited opportunity for role substitution by other crew members.

Each crew member's role is critical as part of the wider team, and there is little opportunity to swap one role for another without compromising safety or operational efficiency. Specifically, the STCW consists of 66 Knowledge, Understanding & Proficiency (KUPs) items across 19 competence themes outlining the minimum standard of competence for officers and ratings across all shipboard areas (Nav, ER, Radio, Cargo).

By establishing these standards, the STCW defines the skills and competencies required to fulfil designated roles effectively. Autonomy will redefine how – and where – many of these roles are delivered. Some will be augmented or superseded by autonomy, while others may be transferred to Remote Operating Centres. It will be important to maintain a human-centred approach and focus on the emergent issues of seafarers' training and competency standards as MASS development continues.

Specialised training programmes will need to be developed to encompass the unique technical, operational, and regulatory facets of MASS, including Bridge Navigation Systems. Traditional training and assessment may require modification or supplementation with innovative methods such as simulation-based assessments or real-time performance monitoring.

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Engagement from stakeholders will be required to review and identify potential gaps or impediments within the Convention. Ethnographic research through engagement, observation and interviews with mariners will be important in identifying the crucial factors influencing crew dynamics, decision-making processes, training needs and safety concerns. By studying the behaviours, practices, and communication patterns of crew members, Human Factors experts will be better able to identify the potential challenges and opportunities associated with autonomous systems in maritime operations, though care must be taken to mitigate the potential for bias that may be caused through subjective assessment dependent on the evaluator's interpretation and skills.

The ultimate goal is to apply autonomy to simplify the burden on the human in various roles, which could lead to an expectation that crew decision making is more likely to involve exceptions or emergency situations. New training will be required to empower people to deal with the more complex technical problems not suited to autonomous systems, whilst recognising that their experience level will almost certainly be reduced to the equivalent role in conventional ship operations.

Close collaboration between regulatory bodies, industry stakeholders, and technical experts will be essential to develop robust regulatory frameworks that help ensure the safe and effective operation of MASS in the maritime domain.



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4.3. Adapting SOLAS for MASS

Seafarers play a critical role in implementing the Safety of Life at Sea (SOLAS) Convention. They are responsible for ensuring compliance with safety procedures, maintaining equipment, responding to emergencies at sea and executing vital tasks such as navigation, firefighting, and lifesaving operations.

Despite advancements in technology, the human element has to date remained indispensable in maritime operations. Humans' judgment, adaptability and problem-solving skills are essential in addressing unpredictable situations and ensuring the safety of crew, passengers, and cargo.

Wherever we sit on the 'sliding scale' from Bridge Navigation Assist through to Full Autonomy, mutual aid at sea, including the rescue of not only crew but also any persons in distress, should continue to be prioritised. Vessels in distress cannot be overlooked, and it is necessary for autonomous systems to play a role in rescue operations wherever possible.

How this is approached and adopted for MASS is a debate requiring the involvement of shipowners, ship management companies, technology providers and regulators as critical stakeholders. How Artificial Intelligence can safely augment or replace certain roles also needs to be demonstrated and regulated.

Interpreting SOLAS in an autonomous environment

Various provisions within SOLAS could create challenges in interpretation in a MASS environment. Examples include:

- **SOLAS Ch. II-2, Reg. 7 Detection and Alarm –** This regulation calls for fire patrols, including regular patrols by the crew to detect and extinguish fires as early as possible, without relying on fire detectors.
- **SOLAS Ch. III, Life Saving Appliances and Arrangements –** In this section, Reg. 20 on Operational readiness, maintenance, and inspection specifies the recording of various life-saving equipment in the Inspection and Logbook. Changing manning levels in an autonomous environment may change the requirements for lifesaving equipment, though lifeboats may also serve as rescue boats, and may be needed to deal with marine casualties.
- **SOLAS Ch. IV, Radiocommunications –** Reg. 12 (Watches) requires ships to maintain a constant radio watch on a range of frequencies. In addition to DCS messages, support for voice communications is also required. Reg. 16 (Radio Personnel) also states that every ship must carry personnel qualified for radio communications and that a crew member must be designated as responsible for distress communications.
- **SOLAS Ch. V, Safe Navigation –** Reg. 5 (Meteorological services and warnings) require meteorological observations to be recorded and communicated to land stations and surrounding vessels in the event of severe weather. In addition, Reg. 24 (Use of heading and/or track control systems) requires that the possibility to switch to manual steering be maintained where heading and/or track control systems are in use, and that manual steering should be tested under the supervision of a responsible officer after prolonged use of HCS/TCS.

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4.4. Integrating autonomous systems with conventional rules

The COLREGs, STCW, and SOLAS are designed with a broad scope, which poses a challenge for machine interpretation due to their inherent ambiguity. While natural language processing (NLP) has advanced considerably in recent years, computers, unlike humans, struggle to infer context and cultural nuance in communication, relying on strict rules and patterns. Human understanding often involves common sense and an appreciation of broader context that is beyond most computational systems available today.

To deal with this language ambiguity, explicit codable requirements and unambiguous terminology are needed, establishing clear definitions and rules that are readable by autonomous systems. Translating regulations into machine-readable formats and decision-making algorithms will require rigorous and well-structured testing and validation. This should involve collaboration between maritime authorities, industry stakeholders and technology experts on an international scale.

In a broader context, we must also recognise that there is a wider group of standard operating procedures used for complying with international rules (conventions, codes, resolutions) that are necessary for the safe operation of ships. Some are documents issued by the IMO, some are working documents prepared by industry associations, and some are internal documents describing established 'acceptable norms.' Some procedures may have been passed down by seafarers from one to the next through on-the-job training as tacit knowledge and are not formally documented. Where appropriate, these procedures need to be clarified and then defined as requirements, so that a functional architecture can be described, which will also help in providing traceability to demonstrate compliance with the rules.

The creation of standardised, codable and machine-readable terminology would enhance safety, foster interoperability and build trust in the maritime industry, ensuring that autonomous systems can interpret and adhere to regulatory requirements accurately. A safety and regulatory compliance initiative of this type would augment, rather than replace, existing rules and would deliver a significant step toward the realisation of MASS potential.

Unfortunately, there may not be time for all of the required documents to be augmented or updated to meet the 2032 mandatory adoption of IMO's MASS Code (see text box in Section 3.1.Legal and ethical questions in MASS development), as ratification and relevant Sub-Committee processes would need to be completed by early 2025. In addition, without structured evidence-based inputs, these changes may suffer similar delivery issues to the main MASS Code itself, with differing levels of subject matter expertise, paper-based inputs and the need to negotiate international agreement likely creating ratification challenges.

Nevertheless, a comprehensive review of STCW is required, involving maritime industry stakeholders, regulatory bodies, technology experts, and academia, to assess its adaptability to autonomous technologies and MASS. This may lead to new training procedures, modules and certification programmes to ensure that personnel are equipped with the necessary skills for MASS operation, maintenance, and oversight. Additionally, clear standards for human-machine interfaces (HMI) for onboard MASS systems should be established, alongside guidelines for remote operations and monitoring, including training requirements for remote operators.

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5. Regulatory sandboxes – a testbed for MASS

A Regulatory sandbox for transportation is a controlled – but experimental – environment where new transportation technologies, services, and business models can be tested under real-world conditions without facing the full burden of existing regulations. Such initiatives have evolved in acknowledgment of the rapid pace of technological advancements such as MASS, with the aim of discovering and removing issues with their integration into existing regulatory frameworks, whilst avoiding the stifling of innovation and maintaining public and operator safety.

This approach serves as a platform for regulators and policymakers to observe the real-world implications of new technologies and services before shaping and formalising the requisite new regulations.

By allowing participants to operate within a controlled environment, regulators can gain insights into the potential impacts on safety, infrastructure, and other relevant factors, enabling them to make informed decisions about gaps or challenges in existing frameworks.

Furthermore, a Regulatory sandbox fosters collaboration and knowledge-sharing among sector participants, including Owners/Operators, Regulators, Class Societies, Government Agencies and Technology Developers. By bringing together diverse perspectives and expertise, this collaborative approach helps to build trust and transparency between innovators and regulators, reducing the likelihood of conflicts or misunderstandings as MASS is introduced to the sector. More widely, it also facilitates discussions on the ethical element of autonomy exploitation, delivering a simplified and understandable model that can be used to develop trust in these new technologies amongst the public and all wider stakeholders.



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Benefits of Regulatory sandboxes for MASS testing

Innovation acceleration: Sandboxes provide a controlled environment for testing and deploying MASS without the burden of strict regulations. This ultimately facilitates accelerated development and deployment of innovative solutions in the maritime sector.

Exemption from certain regulations: Participants in regulatory sandboxes are often granted temporary exemptions from certain regulatory requirements, enabling them to test and experiment with innovations without being immediately subject to the full range of regulatory constraints.

Safety improvements: Sandboxes facilitate the identification and resolution of safety concerns through real-world – and potentially synthetic – testing in a controlled environment. This approach offers the potential to enhance the overall safety of MASS operations by refining technologies and regulatory frameworks based on practical experience.

Limited scope and duration: Regulatory sandboxes are commonly designed to operate for a specified time and within a specific scope. This allows Industry to test innovations and Regulators to monitor and assess the impact of the technologies introduced without compromising the overall regulatory framework.

Controlled environment: The sandbox provides a controlled environment with defined boundaries where the impact of new technologies and/or operating models can be observed and assessed. This helps Regulators understand potential risks and benefits before deciding on the appropriate regulatory approach.

Close regulatory oversight: While there are exemptions, regulatory authorities will still typically monitor and supervise activities within the sandbox. This oversight ensures that the testing is conducted responsibly and that any risks or potential harms are identified and can be promptly addressed.

Risk mitigation strategies: Regulatory sandboxes typically include mechanisms to identify and manage risks associated with the innovations being tested within a controlled environment (before widespread implementation). This may involve setting certain limits on the scale of operations.

Collaboration and feedback: The sandbox model facilitates collaboration between stakeholders including Owners/Operators, Technology Developers, Regulatory Bodies, Class Societies and other stakeholders. This collaborative approach helps Regulators to gain a deeper understanding of the innovations presented and allows industry players to provide feedback that may contribute to the development of informed and adaptive regulatory frameworks.

International standardisation: Where open and transparent, sandboxes facilitate the opportunity to establish international MASS standards as Regulators collaborate and share experiences. Promotion of interoperability and a harmonised approach to regulation on a global scale is also inherent in such an environment.

Learning opportunities: Regulatory sandboxes provide valuable learning opportunities. In creating a platform for shared learning and experience, all parties gain insights into the capabilities and limitations of autonomous systems. Innovators learn about regulatory requirements and potential challenges in a real-world context.

Regulatory adaptation: Regulators gain insights into emerging technologies and their implication through observation of the real-world performance of MASS, allowing regulations to be adjusted based on actual data and outcomes. In a pan-domain context, Sandboxes often promote a more adaptive and responsive regulatory framework that is better able to keep pace with technological advancements.

Market maturation/confidence: Safe and controlled demonstration offers a clear pathway to building confidence in MASS among industry, investors, wider sector stakeholders and the public. Sandboxes encourage the market to embrace new technologies with the assurance that safety and regulatory compliance are being prioritised.

Applicability to novel transport technology solutions: Sandboxes are often established for industries or sectors that are undergoing rapid technological changes or are characterised by high levels of innovation, such as MASS.

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Ultimately, the goal for MASS is to strike a balance between fostering innovation and ensuring safety, and a sandbox can contribute to this goal by providing safety through the control of risk. However, moving the project beyond the sandbox requires the residual risk to be identified and managed – there is no benefit in designing a MASS that can only operate in a sandbox.

Table A (below) describes the main properties of a sandbox, with further granularity exploring the differences between a 'real-world' Physical Sandbox and the Digital Sandboxes that will be required to augment the former – in particular for IMO Level 4 (Full) Autonomy.

The Digital column addresses some of the challenges of the current Physical sandboxes and includes 'in the loop' Virtual Test Environments that provide a scalable and safe platform for extensive verification and validation, reducing the need for costly real-world experiments. This approach is currently being developed as an LR MASS assurance pathway.

Sandbox

Controlled safe experimentation and learning

- Enables entrepreneurial development and informs regulatory policy.
- Disruptive technology is risk-managed.
- Testing autonomy and AI within set safeguards, providing explanation and a body of evidence from learning.
- Testing innovation without legal constraints.
- Identifies additional items to include, rectify or exclude from scope.
- Flexibility to conduct bespoke tests.
- Possibility to replicate functions in a real-life environment.
- Derives requirements.

Physical sandbox

Challenge of EU & UK Regulatory Sandboxes today

- The need to balance liability protection and innovative experimentation is not yet addressed.
- Potential exposure of trade secrets.
- Tested systems could meet regulatory compliance but lead to unforeseen liability risks.
- Different Administrations may derive different ways to comply.
- Scope of testing will be limited, creating inherent unknown risks.
- May not enable a technology to be deployed at its full potential.
- Makes incremental progress with every experiment, but without concluding with a universal case.

Digital sandbox

Synthetic testbed: Universal compliance in a cost-effective manner

- An LR Vision for safe adoption of autonomy and AI.
- Thorough testing in a cost-effective manner.
- All requirements can be considered under a comparatively reduced cost of compliance while meeting EU AI Act provisions.
- All scenarios are simulated, and technology exploited to its full potential.
- Technology can be scoped for the exact reality the asset will experience in all commercial operational cases required.
- Generates traceable and complete body of evidence of the requirements that will be met, and how, enabling the creation of a Universal Safety Case and Assurance Framework

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5.1. International collaboration

Following a coherent international approach in the administration of exploration and Regulatory sandboxes for MASS provides a unique opportunity for cross-border collaboration and knowledge exchange while ensuring safety, efficiency, and sustainability. This type of initiative has the potential to bring together maritime powerhouse nations, leveraging their expertise, resources and best practices to lead the development of more robust and adaptable regulatory frameworks and to pioneer the adoption of MASS at scale.

By establishing common frameworks, where countries collectively address common challenges such as data and cybersecurity risks, those nations could reduce barriers to entry for new technologies and services, encouraging cross-border MASS deployment and interoperability.

However, within this collaboration it is important to recognise that regulatory application may vary among participants, which may require demonstration of different technological capabilities in different territorial waters. Nevertheless, rather than being a constraint on development, these differences could be considered as an opportunity for a project's outcome to be greater than 'the sum of its parts' when compared to single regulator engagement, introducing an additional requirement for adaptability in testing. This shared learning experience would accelerate the pace of innovation in the sector for all involved nations, driving the adoption of safer, more efficient, and more sustainable solutions globally.

International collaboration on Regulatory sandboxes would promote harmonisation and standardisation, which are essential for scaling up emerging technologies such as MASS for deployment across different territorial waters. This may include transfer of knowledge among regulators and create a feedback model that can be delivered into the IMO. A harmonised regulatory environment would not only reduce compliance costs and time challenges for the industry, but also enhance operator and public trust and confidence in these transformative technologies.

Efficiency and cost-effectiveness in the development and deployment of solutions are inherent in this collaborative approach. Through knowledge exchange and capacity building, nations can iteratively refine their regulatory frameworks to keep pace with technological advancements while adapting to sectoral and societal needs. This collaborative environment will not only deliver benefits in the regulatory space but could also potentially facilitate the establishment of an ecosystem of businesses operating within an interconnected marketplace.

By establishing themselves as leaders, the participating nations can attract investment, create high-skilled jobs, and stimulate innovation centred around maritime technologies. Instead of owners/operators and equipment manufacturers/developers having to individually navigate complex and divergent regulatory landscapes in each country where they wish to operate, collaboration can offer a harmonised approach that would streamline the process, reducing bureaucratic hurdles and speeding up potential time to market.

Overall, by promoting collaboration rather than fragmentation, international Regulatory sandboxes can unlock the full potential of connected transportation technologies to deliver societal benefits and drive sustainable economic development – maritime is, after all, an international enterprise, and the benefits will be shared and multiplied if we can move forward together.

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6. Summary and Call to Action

The advent of MASS represents a pivotal moment in maritime history, presenting both challenges and opportunities that demand immediate attention.

As we transition towards this new era, it is imperative that the industry addresses the development of the regulatory and assurance frameworks necessary to secure the operational safety of MASS.

The current standards established by the STCW, COLREGs and SOLAS conventions have provided a solid foundation for maritime safety and operation. However, the introduction of MASS necessitates a thorough review and augmentation of these frameworks to accommodate the unique challenges and requirements posed by autonomy.

As the collective custodians of maritime safety and standards, all stakeholders in the industry must collaborate to successfully adapt legacy frameworks to accommodate the nuances of MASS. This requires a focused effort to identify gaps, develop comprehensive standards, and establish robust assurance mechanisms. It is essential that we pool our expertise and resources to develop regulations and assurance pathways tailored specifically for the safe operation of MASS that are internationally acceptable. This includes establishing protocols for autonomous navigation, collision avoidance and emergency response, among other critical areas.

We propose that a multi-stakeholder collaboration is formed, between governments, as funding partners, Regulators and their recognised organisations in maritime, and key industry stakeholders, including ship operators, ports and other identified participants in the sector. The main purpose of this collaborative effort would be to define the steps necessary to adequately adapt COLREGs, STCW and SOLAS to facilitate the safe operation of MASS, in line with the ongoing advancements in autonomous systems technologies.

Such development could be undertaken within a structure of linked – but independently framed – Regulatory sandboxes, formed to deliver this intended outcome, and working to provide evidence-based results within the timeframe required for integration into the MASS Code.

By working together, the maritime industry can lay the foundation for the safe and efficient integration of MASS into global operations, leveraging our collective expertise to develop and refine the regulations required. This collaborative effort would not only facilitate innovation and technological advancement but also ensure the highest standards of safety, security, and environmental protection – ensuring that the full potential of autonomous shipping in the maritime industry can be realised.

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With a rich history spanning more than 140 years, Mitsui O.S.K. Lines (MOL) has established itself as a leader in maritime transportation, operating a diverse fleet of over 800 vessels including dry bulkers, tankers, Liquid Natural Gas (LNG) carriers and car carriers. The company plays a pivotal role in the global maritime industry through its commitment to safe technology implementation and advancement while maintaining compliance with the most stringent of regulations. Recognising the paramount importance of ensuring the safety of the fleet, crew, cargo, wider maritime stakeholders and the marine environment, MOL is committed to continuous improvement and the adoption of cutting-edge technologies.

Rather than viewing regulations as an impediment, MOL sees them as benchmarks for setting higher safety standards, taking a proactive approach to safe technology introduction and advancement. The company actively aims to collaborate with regulatory bodies, industry partners, and research institutions to develop and implement innovative solutions that not only meet but exceed regulatory requirements, particularly with respect to the safety of its own ships and crew as well as the wider sector. Through this collaborative approach, with open communication channels and engaging in knowledge-sharing initiatives, MOL ensures it remains at the forefront of the dynamic landscape of global shipping, continuously striving to set new standards for excellence and corporate reputation.

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Established in 1760 as the pioneer in the assurance of maritime safety, Lloyd's Register (LR) has evolved into a global leader in maritime technical and engineering professional services. As a trusted advisor to shipowners, operators, and regulatory bodies alike, LR continues to redefine industry standards and drive advancements in maritime safety, sustainability, and efficiency. In its ongoing commitment to technological advancement and adaptability to meet the evolving needs of the industry, LR has consistently demonstrated its ability to embrace emerging innovations and methodologies, ensuring that its classification and certification processes remain robust, reliable, and relevant in an ever-changing landscape.

Through collaboration with industry partners and wider stakeholders, LR continually explores new technologies and best practices to enhance safety, efficiency, and sustainability in maritime operations. Nowhere is this more evident than in its support for autonomous systems development and its dedication to quality, innovation, research and development.

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8. Glossary of Terms

AAVs	Autonomous Aerial Vehicles
AI	Artificial Intelligence
BRM	Bridge Resource Management
CAV	Connected Autonomous Vehicles
COLREGs	International Regulations for Preventing Collisions at Sea 1972
DFFAS	Design of the Future of Full Autonomous Ships
DfT	Department for Transport
HMI	Human-Machine Interfaces
HTW	Human Element, Training and Watchkeeping
ICAO	International Civil Aviation Organisation
IMO	International Maritime Organisation
ISM	International Safety Management
KASS	Korea Autonomous Surface Ship
LNG	Liquid Natural Gas
LR	Lloyd's Register
MASS	Maritime Autonomous Surface Ships
MCA	Maritime and Coastguard Agency
MOL	Mitsui O.S.K. Lines
MPA	Maritime and Port Authority of Singapore
MSC	Maritime Safety Committee
NLP	Natural Language Processing
ODD	Operational Design Domain
OOW	Officers of the Watch
ROCs	Remote Operation Centres
SMS	Safety Management Systems
SOLAS	International Convention for the Safety of Life at Sea
STCW	International Convention on Standards of Training, Certification and Watchkeeping for Seafarers
UNCLOS	United Nations Convention on the Law of the Sea
V2X	Vehicle-to-Everything Communication Protocols

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9. References

1 European Maritime Safety Agency. (2022, September 29). Analysis of navigation accidents. SAFETY4SEA. https://safety4sea.com/emsa-analysis-of-navigation-accidents/

2 International Council on Systems Engineering. (n.d.). What is systems engineering? https://www.incose.org/about-systems-engineering/what-is-systems-engineering#:~:text=Systems%20Engineering%20is%20a%20transdisciplinary,%2C%20technological%2C%20and%20management%20methods

3 Sivori H., Brunton L. (2023). Out of the box – Implementing autonomy and assuring AI in the maritime industry. Thetius & Lloyd's Register. https://www.lr.org/en/knowledge/research-reports/2023/ai-and-autonomy/

4 Edwards, L. (2022). The EU AI Act: A summary of its significance and scope. Newcastle University. https://www.adalovelaceinstitute.org/wp-content/uploads/2022/04/Expert-explainer-The-EU-AI-Act-11-April-2022.pdf

5 International Maritime Organization. (1972). Convention on the international regulations for preventing collisions at sea (COLREGs). https://www.imo.org/en/About/Conventions/Pages/COLREG.aspx

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